Travelling Salesman Problem

```
def tsp(g):
    n = len(g)
    v = [0]
    c = 0
    while len(v) < n:
        last = v[-1]
        nxt = min([i for i in range(n) if i not in v], key=lambda i: g[last][i])
        c += g[last][nxt]
        v.append(nxt)
    c += g[v[-1]][v[0]]
    v.append(v[0])
    return v, c



g = [[0,10,15,20],
     [10,0,35,25],
     [15,35,0,30],
     [20,25,30,0]]

path, cost = tsp(g)
print(path, cost)
```

Output : Tour: [0, 1, 3, 2, 0]

Total Cost: 80